batmonitor 0.58-c: charge path (every 5 s) | blue = pure decision function (bits + state, enumerated by make check) | green = matrix / amounts | [CBMC] = proved for all inputs

inputs
PCC (+ = export), PCC 5 min mean, Sofar Bat1 W (+ = charge), Multis AC-in,
outdoor temperature, BMS SoC / U / CCL, time (local, solar noon)

PCC

export above 18 kW: who takes it (user 2026-10-10: boost and air conditioning before DO4)

order	who	when
1	stacks (D4 export cap, May-Aug)	every 5 s: the export above 18 kW, up to ~12 kW charger
2	R290 boost (r290_boost.py, .218)	cron 1/min: PCC ≥ 18 kW -> hot water 55 deg C
2	attic air conditioning (hantech.py, .218)	cron 1/min: PCC ≥ 18 kW -> on
3	D5 DO4 pulse -> pv_relay/DO4 (FoxESS shedding)	PCC > 20 kW for 2 min (DO4_GRACE_S), at once > 23 kW (DO4_HARD_W); "1", 3 s later "0"; at most every 5 min; batmonitor/do4_last retained
protection rule (0.51-c), separate from the order above		
!	D5 AC-out overfeed -> DO4 at once	only with the grid offline (vebus /Ac/ActiveIn/Connected != 1): PV feeds > 5 kW into one MultiPlus AC-out for 10 s (factor 1.0 rule, 5000 VA per unit; FoxESS on AC-out1 above 15 kW); with the grid there: bypass via the 50 A transfer relay

surplus (chain_mode)
Bat1_eff = Bat1 discharge x 1, Bat1 charge x BAT1_CHARGE_FACTOR 0.5
surplus = z2 + Bat1_eff + own_charge - own_discharge (+ D6 share)
z2 = PCC + WP part the stacks do not cover (S2, as discharge): heat pump matrix below

heat pump matrix BM_WP_RULE (0.58-c: first match top-down; stacks cover min(share x WP, cap), the rest onto z2) - details wp_matrix.pdf

row	when	rule text	share %	cap W	make check
LOWSOC	lowest stack < WP_SOC_MIN 21 % (back ≥ 23 %)	WP:LOWSOC	0	-	221184
FC	good forecast (stacks above target_soc)	WP:FC	100	-	27648
FCBAD	bad forecast (target_soc 100)	WP:FCBAD	WP_BAT_SHARE_BAD 50	-	10368
WINTER	season winter	WP:Z2	0	-	61056
TRANS	season transition	WP:ALL / T1900	100	WP_BAT_MAX_TRANSITION 1900	61056
SUMMER	season summer	WP:ALL	100	-	61056
orange = share 100 %, no cap: **D6 WPPV** allowed (by day, no D4 cap, PV covers house + WP): z2 += min(50 % WP, 2000 W)					

D4 peak window as a forecast export cap (0.49-c: pure function bm_d4)
goal: keep the stacks free for the afternoon peaks, charge only the export above EXPORT_CAP (summer 2026: DO4 on 55 days, 11:45-16:15)

clear-sky model dc(t) (bm_logic.c calc_arrays_kt, from fox2db_logic.h / clearsky.tex)

inputs	
time	unix clock (NTP, Fritzbox)
site	LAT 47.6811 N, LON 11.5732 E (Lenggries)
outdoor temperature	aussen/temp (MQTT, < 15 min old)
steps	
sun	NOAA: elevation, azimuth from time + site
atmosphere	air mass AM = 1 / sin(elev) (≤ 37), Meinel T = 0.7^(AM^0.678)
4 PV strings	tilt / azimuth from south / power, fitted per string (gen_pv_strings.py, 14 clear days): Sofar PV1 west 60 deg / +33 deg / 19430 W, PV2 south 68 deg / -12 deg / 7690 W FoxESS pv2 east 59 deg / -29 deg / 22036 W, pv1 west 67 deg / +32 deg / 2781 W
horizon	trees east: below 23 deg (south-east 15 deg) only 25 % diffuse
month	KT_MONTH empirical: Dec 0.13 ... May 0.91
temperature	cell = outdoor + NOCT 45 x sun; -0.30 %/K above 25 deg C (0.85..1.0)
dc(t) = Σ power x T x kt x cos(incidence) x shade x temperature factor	

dc(t)

forecast scan (ladesperre wrapper, not a decision)
now .. 21:00 in 15 min: PV = clear sky x OWM slot kt x correction,
anchored to the measured Sofar PV of now (fades over 1.5 h)
expected export = PV - HOUSE_EST 1000 W; peak_h, win_end_h (last export > cap)

bm_d4 bits	
enabled	BATMONITOR_LADESPERRE and EXPORT_CAP > 0 (18000 W)
season	May..August
fc_cap_ahead	forecast export > EXPORT_CAP before 21:00
surplus_cap_on / hold	surplus - EXPORT_CAP > 200 W / > 20 W
chg_prev	a stack charged last cycle
new_day, pcc_peak	display: PCC > 20 kW seen today

bm_d4 state: peak_today (display)
cap_active = enabled and season and fc_cap_ahead
charge above the cap = surplus_cap_on or (chg_prev and surplus_cap_hold)
block (wait, LADESPERRE 0 W) = cap_active and not charge above the cap
cap_active -> charge target PCC = +EXPORT_CAP (rule text CAP)
[make check: 384 combinations, 48 cap, 24 waiting; fixed point]

cap_active

replaces 0.48-c latch / ratio / bad weather
(only 12 of 4096 combinations blocked, one cloudy cycle ended the window for the day): the anchored forecast follows clouds within one cycle

block

D6 WPPV (0.56-c, user 2026-10-24): heat pump on PV -> half of it into the stacks, the grid buys it as cheap heat power

bm_d6 bits	
enabled	WP_PV_CHG_SHARE > 0 and WP_PV_GRID_MAX > 0
row_pv	bm_wp_covers_all: row share 100 %, no cap (SUMMER, FC)
day, wp_on	PV ≥ 100 W; heat pump ≥ 300 W, em0 fresh
pcc_fresh, no_cap	PCC there, inverter not stale; D4 export cap not active
surplus_gt_on / ge_hold	surplus without the share > 200 W / ≥ 0 (PV covers house + WP)

bm_d6 state: wp_pv
wp_pv = gate and (surplus_gt_on or (wp_pv and surplus_ge_hold))
share = min(WP x WP_PV_CHG_SHARE 50 %, WP_PV_GRID_MAX 2000 W)
chg_ref and surplus += share -> PCC settles at -share
self-holding: own charge +share, PCC -share, surplus unchanged
[make check: 384 combinations, fixed point; state.json chain.wp_pv_w; rule text WPPV]

surplus + share

chg_ref + share

D2 bm_d2 per stack state: prot, force, low, chg_prev
charge_mode: surplus > 200 W on, held while > 20 W
first match: BMS_MISSING > FORCE (force or low) > STALE > LADESPERRE (charge_mode and block) > CHARGE (SoC < 100) / FULL > PROT > CHARGING > DIS
LOW (0.52-c, state low): on at DCL < DCL_FORCE_A 1 A or lowest cell < CELL_FORCE_V, off when DCL is back, the cell ≥ CELL_FORCE_RELEASE_V 3.25 V and (0.55-c) SoC ≥ SOC_MIN_RELEASE; LOW -> FORCE (charge, no discharge); Lenggries env: CELL_FORCE_V 3.05 V
[make check, CBMC h_d2_bits]
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5760 · STALE 1152 · LADESPERRE 288 · CHARGE 216 · PROT 216 · CHARGING 180 · DIS 180 · FULL 72

mode per stack

mode matrix BM_MODE_RULE[mode] (0.48-c): what each mode means for the setpoint

mode	action	capped	rule text
BMS_MISSING	0 W	-	NAME
FORCE	FORCE_CHARGE_W 500	**1** (0.48-c)	FORCE(soc %) [LOW]
STALE	0 W	-	NAME
LADESPERRE	0 W	-	LADESPERRE(peak h)
CHARGE	charge amount	1	CHARGE WP: SRC:
FULL	0 W	-	NAME
PROT	0 W	-	PROT(soc %)
CHARGING	0 W	-	CHARGING(BMS W)
DIS	discharge amount	1	(S4: PROP / IDLE)

chg_cap per phase (one per stack)
min(CHARGER_A 70 A x U / 0.93, CHARGE_MAX_PHASE 4900, BMS CCL)
~4.0-4.2 kW per phase -> ~12 kW for the three Multis

CHARGE

FORCE

charge amount (chain_charge, whole watts since 0.48-c)
want = max(0, own_charge + KP_PM 1010 x (chg_ref + Bat1_eff - target) / 1000)
chg_ref = z2 + D6 share (0.56-c)
target = SOYO_TARGET, + EXPORT_CAP while D4 cap_active (0.49-c)
at most the sum of chg_cap over the charging phases
[CBMC h_charge]

FORCE: min(FORCE_CHARGE_W, chg_cap) per phase
from the grid, obeys the BMS CCL (was: 500 W even at CCL 0)
at any time of day: no TOU window (removed 2026-10-11)

0 W rows

alloc_levels (0.48-c: one split core for charge and discharge)
charge: one level per stack in CHARGE_PRIORITY order (Stack1, Stack2), the lead stack last, each up to min(chg_cap, CHARGER_CAP_PHASE); water filling within a level;
only what the caps cut goes on; a rest up to CHARGE_MAX_PHASE on all charging phases
[CBMC h_charge without bank assumption, h_alloc_discharge]

setpoints per phase (+ = charge) -> Multis
rule text: CHARGE WP:.. SRC:.. BAL/EQ [CAP] [WPPV] [CCL] | LADESPERRE(h) | FULL | FORCE(soc) [LOW]

clear summer day: how the export cap keeps the PCC below 20 kW			
time	**PV / PCC**	**batmonitor**	**stacks**
morning	export below the cap	LADESPERRE (wait)	0 W, stay free
midday .. afternoon	export above 18 kW	CHARGE CAP: PCC held at 18 kW	take the export above the cap
stacks at their charger limit	export still above 18 kW	R290 boost + air conditioning; DO4 after 2 min > 20 kW	full charge power
after the forecast window	export below the cap again	CHARGE (no cap)	fill up to 100 %
cloudy forecast	no export > cap expected	no cap	charge at once
clouds come up	measured PV low	anchor drops -> cap ends	charge at once